

Report: Password & Account Policies

1. Introduction

Password and account policy configuration was performed to strengthen authentication management within the Active Directory domain. The default domain password policy was reviewed using Group Policy Management Console to verify parameters such as minimum length, complexity, history, and expiration. PowerShell was used to identify accounts with the “Password Never Expires” flag enabled and to check for any user passwords older than 90 days. A test account was selected, and a password reset was applied with the enforcement of change at next logon. All actions were performed on the domain controller and verified using ADUC and PowerShell output.

2. Objective

The objective of this task set was to perform account-level policy checks and apply secure password control settings. The domain password policy was examined for strength and compliance. Accounts with non-expiring passwords were detected using PowerShell. An automated check for accounts with passwords older than 90 days was executed. A test account password reset was performed and enforced at login to validate administrative control and policy enforcement. All results were verified with administrative tools to confirm proper application.

3. Tools used

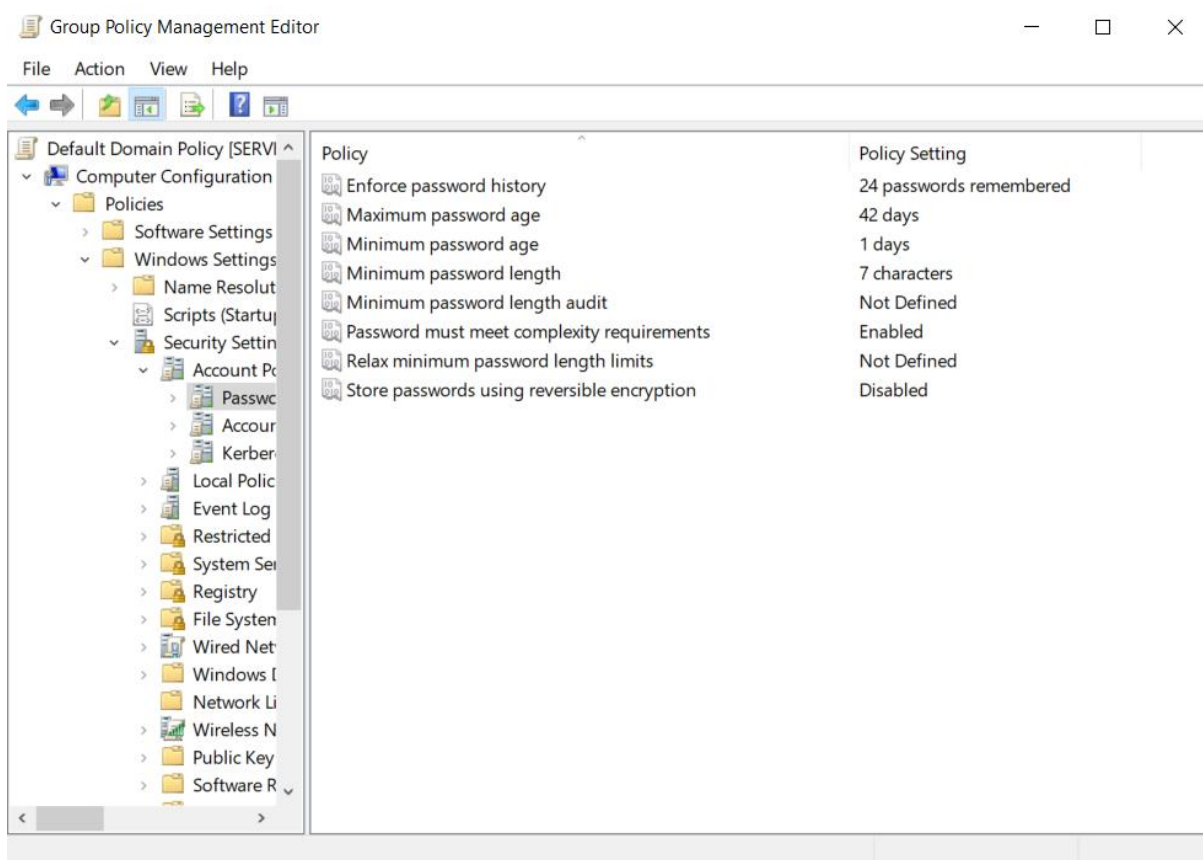
- Windows Server (AD DS)
- Group Policy Management Console (GPMC)
- Windows PowerShell (ActiveDirectory Module)
- Active Directory Users and Computers (ADUC)

4. Methodology

4.1 Review Default Domain Password Policy

Group Policy Management Console was used to open and review the Default Domain Policy. The policy was located under Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Password Policy. The following values were recorded:

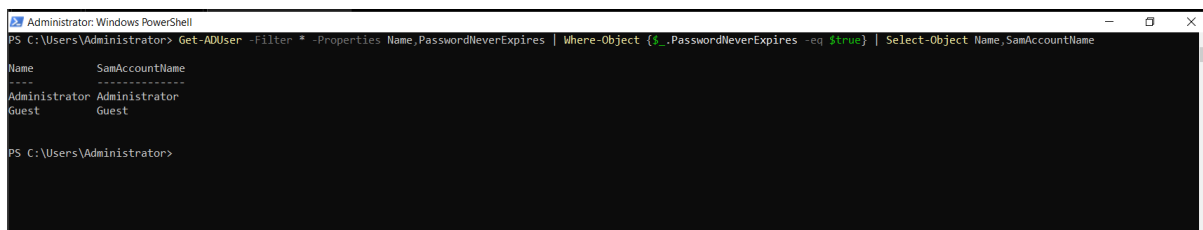
- Minimum password length: 7
- Maximum password age: 42 days
- Minimum password age: 1 day
- Enforce password history: 24
- Complexity requirements: Enabled
- Reversible encryption: Disabled



(Default Domain Password Policy)

4.2 Identify Accounts with “Password Never Expires”

PowerShell was used with the command `Get-ADUser -Filter * -Properties PasswordNeverExpires` to identify user accounts where the PasswordNeverExpires flag was set to True. The Administrator and Guest accounts were returned as having this setting enabled.



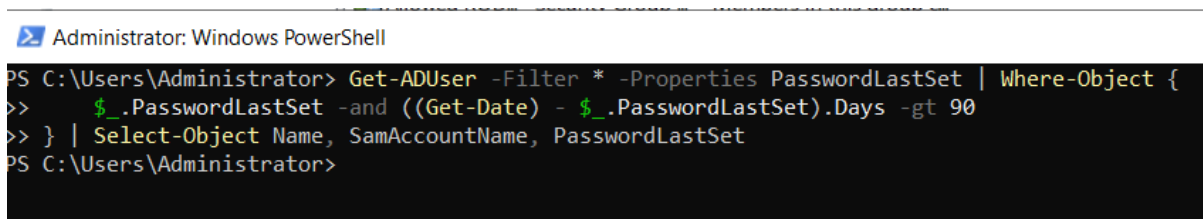
```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADUser -Filter * -Properties Name, PasswordNeverExpires | Where-Object {$_.PasswordNeverExpires -eq $true} | Select-Object Name, SamAccountName
Name                SamAccountName
-----
Administrator       Administrator
Guest                Guest

PS C:\Users\Administrator>
```

(Accounts with "Password Never Expires")

4.3 Identify Users with Passwords Older Than 90 Days

A PowerShell command was executed to identify users with a PasswordLastSet date exceeding 90 days. A filter condition was used to exclude accounts without this property. The output indicated that no user accounts had passwords older than 90 days at the time of checking.



```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADUser -Filter * -Properties PasswordLastSet | Where-Object {
>>   $_.PasswordLastSet -and ((Get-Date) - $_.PasswordLastSet).Days -gt 90
>> } | Select-Object Name, SamAccountName, PasswordLastSet
PS C:\Users\Administrator>
```

(users whose passwords haven't been changed in over 90 days- none)

4.4 Force Password Reset for a Test User

The password for Aarav.Finance was reset using Set-ADAccountPassword. The ChangePasswordAtLogon flag was set to \$true using Set-ADUser to enforce a password change upon next login. The settings were verified through the ADUC Account tab.

Aarav.Finance Properties?

×

Member Of

Dial-in

Environment

Sessions

Remote control

Remote Desktop Services Profile

COM+

General

Address

Account

Profile

Telephones

Organization

User logon name:

Aarav.Finance

@yourdomain.local

User logon name (pre-Windows 2000):

TASK\

Aarav.Finance

Logon Hours...

Log On To...

☐ Unlock account

Account options:

☐ User must change password at next logon

☐ User cannot change password

☐ Password never expires

☐ Store password using reversible encryption

Account expires

☒ Never

☐ End of:

Monday

,

June

9, 2025

OK

Cancel

Apply

Help

(Before giving command to change password at login)

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Set-ADAccountPassword -Identity "Aarav.Finance" -Reset -NewPassword (ConvertTo-SecureString
"Temp@Reset123" -AsPlainText -Force)
>>
>> Set-ADUser -Identity "Aarav.Finance" -ChangePasswordAtLogon $true
>>
PS C:\Users\Administrator>
```

(Powershell command to force user to change password at new login (Temp@Reset123)
(Temp@123))

Aarav.Finance Properties

Member Of	Dial-in	Environment	Sessions
Remote control	Remote Desktop Services Profile		COM+
General	Address	Account	Profile
Telephones		Organization	

User logon name:
 @yourdomain.local

User logon name (pre-Windows 2000):

☐ Unlock account

Account options:

- ☒ User must change password at next login
- ☐ User cannot change password
- ☐ Password never expires
- ☐ Store password using reversible encryption

Account expires

☒ Never

☐ End of:

(After)

5. Recommendations

- Set clear password expiration timelines and notify users before expiry.
- Avoid using the “Password Never Expires” setting unless required for specific service accounts.
- Periodically review password age to ensure compliance with expiration policies.
- Enforce password reset at next login for all newly created or temporary credentials.
- Use scripts to automate checks for password expiration and aging.
- Limit privileged account exclusions and log all exceptions to policy.

6. Conclusion

The completed tasks ensured a baseline of password security was implemented and enforced within the Active Directory domain. Password policies were reviewed and confirmed as active, user accounts with non-expiring credentials were flagged, and outdated password checks were conducted to identify risks. A test password reset with enforcement was successfully executed, validating administrative control over user credential lifecycle. The domain password policy was reviewed through Group Policy Management. A password reset was applied to the Aarav.Finance account, and the option to force a password change at next login was enabled. All results were confirmed using ADUC and PowerShell.